Project Research Design



ACE

CYBER R&D Program

Conversational AI Interface for
Analyst-Driven Threat Investigation

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1.0 Research Method Selection

Since this research investigates how a Conversational AI interface for Elasticsearch can enhance SOC analysts' detection rates and speeds, it is applied in nature, focusing on solving a real-world problem in cybersecurity operations. Additionally, because the study tests a specific hypothesis using a controlled system to determine whether the proposed solution produces measurable improvements, it is also considered an experimental study.

2.0 Research Design Overview

What are you studying?

- This research studies the effects of a Conversational AI interface on SOC Elasticsearch analysts.

What are the key variables or factors?

- The key independent variables in this study include the threat hunting task condition and the query method. The threat hunting task condition refers to the specific cybersecurity scenario being tested, such as detecting failed login attempts or identifying suspicious IP addresses. The query method represents whether the SOC analyst performs threat hunting using traditional manual queries through a SIEM interface or through the Conversational AI interface using natural language input. Comparing these two approaches will reveal the impact of AI-assisted interaction on analyst performance and efficiency.

  The key dependent variables include query latency, threat detection accuracy, and early detection rate. Query latency measures the average time required for a query to return valid results, providing a direct indicator of efficiency. Threat detection accuracy represents the percentage of correctly identified threats compared to a verified ground-truth dataset, reflecting the reliability of each method. Early detection rate refines the concept of early detection by measuring the time

difference between when a threat is first identified by the AI versus manual detection, quantifying the speed advantage of AI assistance.

What is your overall approach (quantitative, qualitative, or mixed)?

- The approach will be quantitative. The research's main goal is to measure query latency, early threat detection rate, and threat detection accuracy.

What do you expect to learn or demonstrate?

- The research expects to demonstrate the effectiveness of AI assistance in SOC environments.

3.0 Research Objectives & Hypothesis

3.1 Primary Objectives

Build a functional Proof of Concept (PoC) Conversational AI interface that supports natural language queries for SOC log data.
Integrate the interface with Elasticsearch to retrieve and display relevant log data.
Support at least two common threat-hunting tasks (e.g., failed logins, suspicious IPs) while measuring query latency, detection accuracy, and analyst satisfaction compared to manual methods.

3.2 Hypothesis

If a Conversational AI interface is developed and integrated with Elasticsearch to support natural-language threat-hunting queries, then, when evaluated against a pre-tagged dataset containing failed-login and lateral-movement events, it will correctly identify at least 95% of those threat-tagged log events and will achieve an F1 score of at least 0.90 across all evaluated threat-hunting tasks.

4.0 Planned Research Steps / Experimentation Plan

Step	Description	Expected Output	Timeframe
1	Define and refine the research question	Project Charter	Week 1-2
2	Conduct a literature review	Literature Review map	Week 3
3	Finalize Research question/hypothesis	Question/Hypothesis	Week 4

Step	Description	Expected Output	Timeframe
4	Create Research Design Outline	Research Design Outline	Week 5-6
5	Develop and configure an experimental testbed	Testbed Setup and Report	Week 7-8
6	Generate or acquire synthetic datasets	Data Acquisition Report	Week 9
7	Execute experiments, collect logs	Raw & Processed Data Logs	Week 10-12
8	Analyze and interpret results	Results & Analysis Summary	Week 13-14
9	Prepare report & presentation	Final Report & Presentation	Week 15-16

5.0 Data Collection Plan

Description	Data Source	Type	Collection Method	Tool
Analyst Perception: Collect at least 10–15 responses on usability, trust, and perceived accuracy after interacting with the AI system	SOC Analysts	Qualitative	Post-task online survey	Google Form
Query Latency: Measure the average time between AI query submission and Elasticsearch response	AI System Logs	Quantitative	Automated timestamp logs	Python script
Threat Detection Accuracy: Compare AI-identified threats to the pre-tagged dataset	AI Query Results	Quantitative	Automated logs	Python script
Early Detection Rate: Track how often AI queries identify threats before manual methods	AI Query Results	Quantitative	Automated timestamp logs	Python Script

6.0 Data Analysis Plan

What statistical tests, visualizations, or coding methods will you use?

- The research will perform statistical tests by computing the means, medians, and standard deviations for query latency, detection accuracy, and analyst satisfaction scores across both conditions (manual vs. AI-assisted). The research will use a paired-sample t-test to determine whether differences in mean latency and accuracy are statistically significant and use Pearson's correlation to examine relationships between satisfaction, accuracy, and latency.

How will you interpret qualitative responses (e.g., coding themes, content analysis)?

- The study will use thematic coding to identify recurring ideas about usability, trust, and clarity of the AI's responses. Furthermore, the study is looking into using NVivo or Python NLP libraries (spaCy, NLTK) for semi-automated keyword extraction.

How will success be measured (KPIs, accuracy, latency, efficiency, satisfaction, etc.)?

- The proof of concept will be considered successful if it produces a functional Conversational AI prototype that enables SOC analysts to query Elasticsearch logs in natural language and retrieve meaningful results for common threat-hunting tasks (e.g., failed logins, suspicious IPs). The system must integrate seamlessly with Elasticsearch for real-time log retrieval and support at least two core adversarial detection tasks. Success will be evaluated using the following KPIs: query latency below five seconds, threat detection accuracy above 90%, user satisfaction scores above 3.5/5 on the post-use survey, early detection of at least 80% of unique threat indicators before human analysts, and an error rate below 5% of total test queries.

7.0 Research Tools, Platforms, and Resources

Tool / Resource	Purpose	Access Needed?	Notes
Elasticsearch	Data analysis	Yes	Provided by ACE Lab
spaCy	Natural Language Processing	No	Install locally
H2OGPT	Interpreting Queries	No	Installed locally
Python Environment	Experiment	No	Installed locally

Tool / Resource	Purpose	Access Needed?	Notes
	scripting		

8.0 Risks & Mitigations

Risk	Potential Impact	Mitigation Strategy
Limited data diversity	May bias results	Use multiple sample sources
Tool access delay	Timeline impact	Request early access
Very short timeframe	Unfinished work	Stay on top of weekly hours, ask for help when needed

9.0 Ethical & Data Handling Considerations

Will your study involve human participants (e.g., surveys, interviews)?

- Yes, the study will involve a small number of SOC analysts or cybersecurity students to evaluate usability, accuracy, and satisfaction when using the conversational AI interface versus manual querying. Participation will be voluntary, and no personal or identifiable data will be collected.

Will you handle sensitive or anonymized data?

- The study will not handle real or classified operational data. Instead, it will use anonymized or fully synthetic SOC log datasets designed to simulate realistic cybersecurity events, ensuring compliance with FERPA and DoD data protection standards

How will you ensure compliance with ethical guidelines, FERPA, or DoD data protection rules?

- The research will comply with ethical, FERPA, and DoD data protection guidelines by exclusively using anonymized or synthetic log datasets within a controlled, simulated SOC environment, ensuring no classified or personally identifiable information is handled. Access will be restricted to approved university and DoD mentors, and all procedures will follow institutional and DoD cybersecurity research protocols to maintain data integrity and confidentiality

10.0 Expected Outputs

Deliverable	Description	Target Date
Prototype	Functional conversational AI POC	November 5th
Data Analysis Report	Summary of results	December 3rd
Final Presentation	15-20 minutes	December 12th

11.0 Change Log

Ver.	Date	Description of Change	Author
v1.1	2025-10-08	Filled out document (first time)	Koi McFarland
v1.2	2025-11-19	Updated Formatting	Koi McFarland

12.0 Mentor Review / Approval

Reviewer	Role	Date	Notes
Student Peer	PI	2025-12-17	Final Deliverable
Faculty Advisor			
DoD/ACE Mentor			